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Business Case 3
Group 12

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1. Introduction and Objective

The objective of this project is to replicate a complex target portfolio using a limited universe of liquid futures contracts. The exercise follows the logic of investment replication: the target portfolio is treated as a black box, since we observe its historical returns but not its exact holdings, trading rules, or internal strategies. The aim is therefore to infer its main risk drivers from observed performance and to build a tradable proxy using liquid instruments.

2. Data Preprocessing

The dataset consists of weekly Bloomberg observations from October 2007 to April 2021. It includes four reference indices and eleven futures contracts across equity, fixed income, commodities, and short-term rates. The raw tickers were mapped into readable asset names and the dataset was checked for missing and stale observations.
No missing values were detected, which allowed us to compute returns and estimate the models without introducing interpolation or imputation assumptions. However, the Lead Future exhibited a very high stale-observation ratio, with roughly 73% of prices unchanged from one week to the next. From a replication perspective, this is important because a highly stale series may contain limited information and may generate misleading correlations. For this reason, its contribution must be interpreted cautiously.

3. Exploratory Data Analysis

3.1. Distributions of the Primary Target Indices

An exploratory data analysis was performed to define the properties of the four primary target indices. The cumulative performance visualization highlighted the divergent behaviors of the constituent asset classes across the observation window. Weekly return vectors, defined as $R_t = (P_t - P_{t-1})/P_{t-1}$, were extracted to compute cross-sectional risk-return metrics.
The broad equity benchmarks (MSCI World and MSCI ACWI) dictate the upper bound of the return profile. They generated annualized yields of approximately 5.9%, but at the expense of substantial volatility, roughly 17.5%, and severe maximum drawdowns exceeding -57%. Conversely, the fixed-income constituent, Bloomberg Global Aggregate, acted as a stabilizing anchor, yielding the highest risk-adjusted performance with a Sharpe ratio of 0.63.

The hedge fund proxy, HFRX Global Hedge Fund, exhibited an asymmetric return distribution. It was characterized by a pronounced negative skewness of -2.30 and an extreme kurtosis of 13.70. This statistical profile indicates that, while the hedge fund component maintains a low annualized volatility of 4.78% in normal regimes, it remains exposed to left-tail realizations during periods of acute market stress.

3.2. Heterogeneity of the Predictor Universe

A parallel exploratory analysis was conducted on the eleven candidate futures contracts to evaluate their standalone distributional properties. Using weekly log-returns, we computed annualized return (μ_{ann}), annualized volatility (σ_{ann}), Sharpe and Calmar ratios, maximum drawdown, higher-order moments, and tail risk indicators, such as 95% Value at Risk and Expected Shortfall.

The empirical results revealed a heterogeneous investment universe. Equity index futures, notably the Nasdaq 100 (*NQ1 Comdty*) and S&P 500 (*ES1 Comdty*), exhibited superior annualized returns, 15.46% and 8.83%, respectively, but were coupled with substantial volatility in the 17%–19% range. On the opposite end of the spectrum, short-term fixed-income futures demonstrated high stability, with annualized volatilities approaching 1%. Commodity contracts, particularly Brent Crude (*CO1 Comdty*) and Lead (*LLL1 Comdty*), presented severe tail risks, evidenced by maximum drawdowns exceeding -64% and extreme leptokurtic behavior, with kurtosis above 16. This heterogeneity confirms that, while the futures universe provides diverse and liquid factor exposures, these instruments introduce substantial idiosyncratic volatility and non-normal tail behavior that must be controlled during the portfolio synthesis.

3.3. Correlation Structure

The correlation analysis provided a first indication of the dependence structure between the target indices and the futures universe. A key result is that MSCI World and MSCI ACWI are almost perfectly correlated. This is expected, since both indices represent broad global equity exposure and therefore react to very similar market drivers. From a replication perspective, this means that the two indices provide nearly the same information: they mainly confirm the dominance of the global equity factor rather than adding two independent sources of risk.

The correlation between the target indices and the candidate futures is also informative. Equity futures are the instruments most strongly related to the equity benchmarks and also display a meaningful relationship with the hedge fund proxy. This suggests that part of the apparent complexity of the HFRX index is linked to traditional systematic market risk, especially global equity risk. The correlation analysis therefore supports the need for a multi-factor replica, where several liquid futures are combined to capture the systematic component of the target while reducing idiosyncratic noise and tracking error.

4. Target Construction and Statistical Diagnostics

4.1. Defining the Synthetic Target

A synthetic target portfolio was engineered by aggregating three specific asset classes: a 70% allocation to the HFRX Global Hedge Fund Index, complemented by a 15% allocation to global equities (MSCI ACWI) and a 15% allocation to global fixed income (Bloomberg Global Aggregate). The target return is defined as:

$$R_{target,t} = 0.70 R_{HFRX,t} + 0.15 R_{ACWI,t} + 0.15 R_{AGG,t} \tag{1}$$

Empirical analysis of this newly minted benchmark reveals a conservative baseline risk profile characterized by low annualized volatility (5.42%) and moderate returns (1.68%). However, like its dominant hedge fund constituent, it preserves severe asymmetric tail risks: negative skewness of -1.80, a high kurtosis of 9.58, and a maximum drawdown of -27.31%. Severe weekly losses occur during systemic crises, such as a -5.73% drop during the COVID-19 shock in March 2020 and a -4.58% drop during the 2008 Global Financial Crisis.

To assess the feasibility of liquid replication, Pearson correlation coefficients (ρ) were computed between the target and the candidate futures. The synthetic index demonstrated the highest codependence with S&P 500 ($\rho = 0.744$), Nasdaq 100 ($\rho = 0.663$), and Euro Stoxx 50 ($\rho = 0.637$). While these global equity factors capture a significant proportion of the target's variance, proving that alternative investments share systematic equity beta, the correlations remain strictly sub-unitary. This mathematically invalidates single-instrument replication strategies.

4.2. Multicollinearity, Autocorrelation, and Temporal Instability

Quantile-Quantile (QQ) plots exposed pronounced deviations from theoretical normality at the extreme tails for both the synthetic composite and its primary predictors. Scatter plot evaluations confirmed a positive but noisy linear codependence, indicating that extreme market realizations may easily disrupt static linear replication.
To assess multicollinearity within the predictor set, Variance Inflation Factor (VIF) metrics were computed. The analysis detected severe collinearity within the US equity complex, with the S&P 500 and Nasdaq 100 futures registering elevated VIF scores of 8.97 and 6.16. To ensure coefficient stability, we executed a deterministic feature selection: the Nasdaq 100 future was eliminated to strip out redundant risk exposures, and the Euro-Bund future (*RX1 Comdty*) was discarded due to its near-zero correlation with the target. The refined design matrix thus comprises nine distinct, liquid contracts.
Autocorrelation Function (ACF) diagnostics, validated via the Ljung-Box Q-test, revealed that the hedge fund proxy and the synthetic target exhibit statistically significant serial correlation ($p < 0.01$). Additionally, Engle's ARCH Lagrange Multiplier (LM) test rejected the null hypothesis of homoscedasticity across the entire asset universe ($p < 0.001$), confirming the presence of volatility clustering. Finally, 52-week rolling window analyses confirmed temporal instability in both rolling correlation (ρ_t) and rolling beta. This violates the assumptions of static Ordinary Least Squares (OLS) estimation, demanding adaptive econometric techniques.

5. Comparative Modeling and Replication Frameworks

To establish a comparison, we tested four replication methodologies. An Exponentially Weighted Moving Average (EWMA) algorithm continuously computes the 1-month 1% Value at Risk (VaR). Should this metric breach an 8.0% threshold, the optimal weights are scaled down to enforce structural deleveraging.

5.1. Rolling OLS Regression

To establish a baseline, a rolling OLS regression was implemented using historical windows of 52, 104, and 156 weeks. The objective function minimizes the sum of squared tracking errors. The 156-week window was identified as the optimal hyperparameter, achieving an out-of-sample annualized tracking error of 3.20%. While this model reaches a high correlation ($\rho \approx 0.749$) and matches the target's central tendencies (annualized return of 3.04% vs. 2.83%), its temporal trajectory reveals parameter instability. It exploits correlated but noisy signals, driving excessive average gross exposure (≈ 1.24) and generating highly erratic portfolio weights.

5.2. Elastic Net Regularization

To address this coefficient instability, an Elastic Net regression was deployed. By incorporating both L_1 (Lasso) and L_2 (Ridge) penalties into the objective function, the model penalizes large weights and mitigates the effects of correlated predictors. Hyperparameter tuning via Optuna identified an optimal configuration utilizing a 156-week rolling window, a high L_1 ratio (indicating a strong feature-selection Lasso effect), and a regularization parameter $\alpha \approx 0.0001$.
This regularization stabilized weights and compressed average gross exposure from 1.24 down to 0.215288. The risk-managed portfolio maintained strict compliance with the 8.0% VaR threshold (averaging a VaR of 1.98%). However, this aggressive shrinkage induced significant estimation bias. By suppressing the weights, the model became conservative, resulting in a degraded tracking error and a reduction in correlation. Given that the design matrix comprises instruments with relatively low remaining multicollinearity (maximum VIF < 3.2), the Elastic Net penalty introduced excessive bias without a commensurate reduction in estimation variance.

5.3. Kalman Filter State-Space Modeling

A Kalman Filter state-space model was implemented. This methodology models the unobservable portfolio weights as a hidden random walk ($\beta_t = \beta_{t-1} + w_t$), recursively updated via Bayesian inference upon observing new market returns. By precisely tuning the process noise covariance matrix (Q) and observation noise scalar (R), the filter achieves a balance between parameter adaptability and noise rejection.
Hyperparameter optimization via Optuna yielded an optimal configuration ($Q = 0.00088$, $R = 0.099768$), indicating a statistical preference for slow, persistent weight evolution while heavily discounting weekly measurement noise. The Kalman Filter outperformed both static benchmarks, minimizing the annualized tracking error to 3.43% and maximizing the full-sample correlation to 0.7628. It completely resolved the leverage problem: the average gross exposure was constrained to a tradable 0.7517, eliminating the erratic coefficient oscillation of

OLS. The embedded EWMA VaR constraint remained entirely unbreached.

5.4. Deep Learning: LSTM Neural Network

To explore non-linear market dynamics, a Deep Learning architecture utilizing a Long Short-Term Memory (LSTM) network was trained using $\mathcal{L} = \sqrt{\text{Var}(R_{replica,t+1} - R_{target,t+1})}$ as a custom Tracking Error Volatility (TEV) loss function, combined with a Mean Absolute Deviation (MAD) penalty to attenuate sensitivity to outlier shocks.
Training was restricted to the first 70% of the time series, with performance evaluated on the unseen 30% test set. Hyperparameter optimization converged on a 61-week lookback window and dual hidden layers of 64 units. Despite achieving stability over 200 epochs, the LSTM's out-of-sample tracking error remained at 3.79%, underperforming the Kalman Filter. The relatively low signal-to-noise ratio inherent to weekly financial data caused the highly parameterized model to struggle to generalize effectively.

5.5. Common Window Comparison

To ensure rigorous comparison, all models are evaluated over an identical out-of-sample period (starting 2017-06-20). This prevents conflating model quality with varying market regimes caused by the rolling models' 156-week training requirement.
On this common window, OLS serves as a strong linear baseline, achieving a 4.01% tracking error and the least negative information ratio (-0.131) due to effective prior feature selection. Conversely, Elastic Net exhibits the worst information ratio (-0.466), as aggressive L_1 shrinkage introduces estimation bias in a low-multicollinearity setting. The Kalman Filter delivers the optimal balance of tracking error (3.79%) and correlation ($\rho = 0.752$), confirming the superiority of dynamic weight estimation for non-stationary targets. Finally, while the LSTM attains the highest correlation (0.762), it yields the lowest information ratio (-0.629): it captures directional co-movement but systematically underestimates returns, reflecting the low signal-to-noise ratio of weekly financial data.

6. Final Interpretation

This project demonstrates that complex, alternative-investment targets can be partially decomposed into liquid market factors. While imperfectly replicable, a significant portion of the target's performance stems from systematic risk—particularly global equity—making futures-based replication valuable for hedging and monitoring opaque portfolios.
The residual tracking error remains highly informative. It represents the genuinely difficult-to-replicate components of the strategy, such as illiquid assets, non-linear exposures, hedge-fund-specific trades, or dynamic allocations.
Among the evaluated models, the Kalman Filter emerges as the most suitable approach. Its dynamic weights realistically adapt to evolving portfolio exposures, offering greater flexibility than static regressions and higher interpretability than neural networks.
Ultimately, the synthetic clone is not a perfect substitute, but a tradable, risk-controlled proxy. It successfully fulfills the core practical role of investment replication: isolating explainable, hedgeable systematic risks from what remains genuinely alternative.